

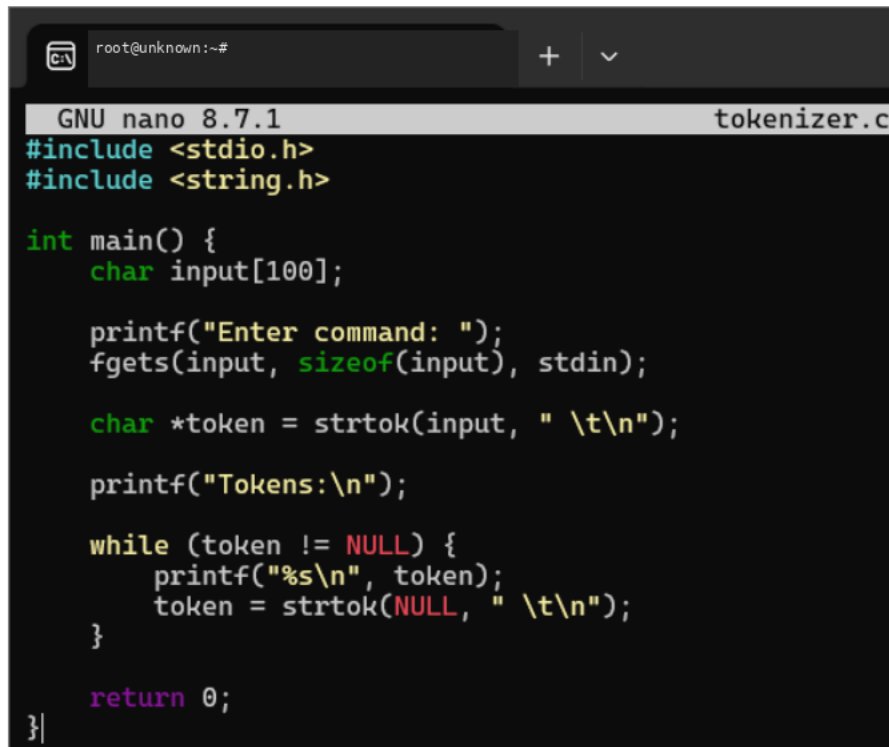
OSSP SKILL

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To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

(a) Code:



```
root@unknown:~#
GNU nano 8.7.1 tokenizer.c
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

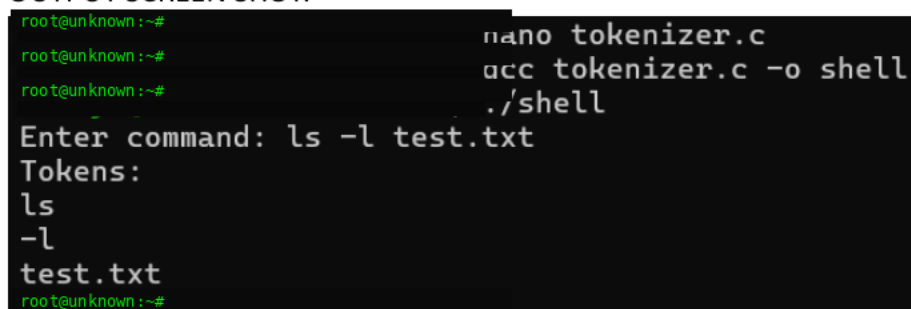
    char *token = strtok(input, " \\t\\n");

    printf("Tokens:\\n");

    while (token != NULL) {
        printf("%s\\n", token);
        token = strtok(NULL, " \\t\\n");
    }

    return 0;
}
```

OUTPUT SCREEN SHOT:



```
root@unknown:~# nano tokenizer.c
root@unknown:~# gcc tokenizer.c -o shell
root@unknown:~# ./shell
Enter command: ls -l test.txt
Tokens:
ls
-l
test.txt
root@unknown:~#
```

(b) Code:

```
GNU nano 8.7.1                                     parser.c *
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0) {
        printf("Error: Empty command\n");
        return 0;
    }

    if (strcmp(input, "ls") == 0) {
        printf("Valid command: ls\n");
    }
    else if (strcmp(input, "pwd") == 0) {
        printf("Valid command: pwd\n");
    }
    else if (strcmp(input, "exit") == 0) {
        printf("Valid command: exit\n");
    }
    else {
        printf("Error: Invalid command\n");
    }

    return 0;
}
```

OUTPUT SCREENSHOT:

```
root@unknown:~# nano parser.c
root@unknown:~# gcc parser.c -o parser
root@unknown:~# ./parser
Enter command: ls
Valid command: ls
root@unknown:~# gcc parser.c -o parser
root@unknown:~# ./parser
Enter command: hello
Error: Invalid command
root@unknown:~#
```